

Outset · AI in Healthcare · Day 3 · Group 1

Predicting good CRISPR guides

Can a model read 20 letters of DNA and guess how well the molecular scissors will cut?

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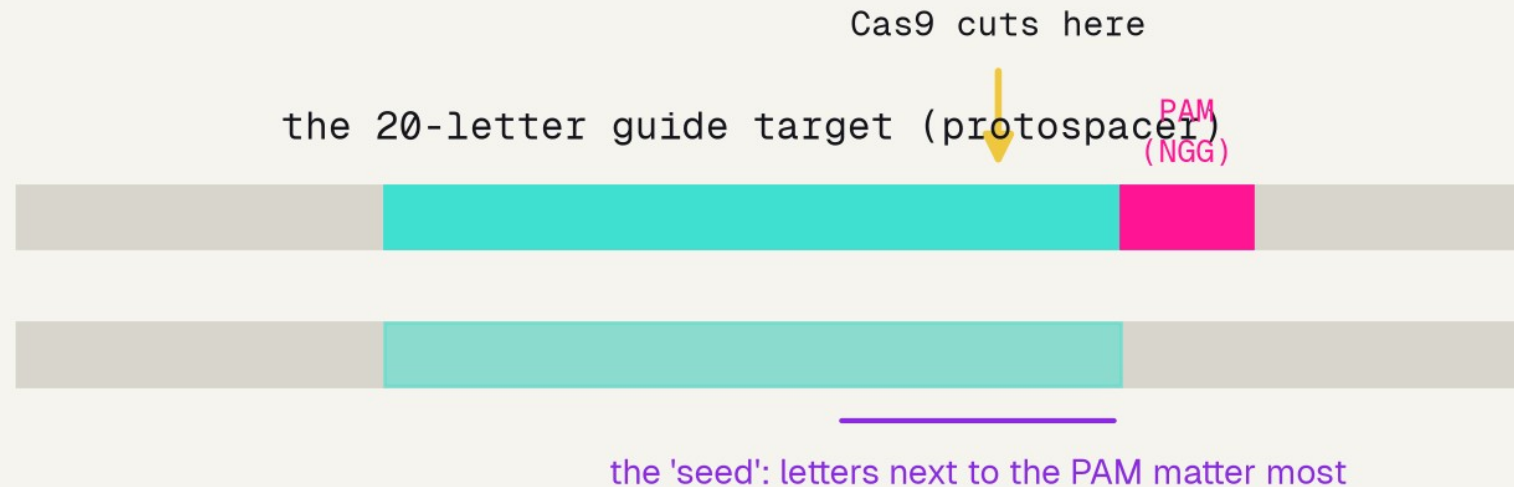
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BACKGROUND

What CRISPR actually is

CRISPR-Cas9 is a pair of molecular scissors you aim at one spot in DNA using a short 20-letter guide (the letters are A, C, G, T). Where the guide matches, the scissors cut -- and that cut turns a gene off or repairs a mutation.



adapted from Hsu, Lander & Zhang 2014

The 20 letters in our dataset ARE this guide sequence. Cas9 locks onto the PAM (the NGG landmark), then cuts a few letters upstream.

How real scientists pick a guide

Two guides that look almost identical can behave completely differently -- one cutting beautifully, the other barely at all. Testing each one in the lab takes weeks, so scientists predict cutting efficiency from the 20 letters instead of measuring it.

SCORE, DO NOT EYEBALL

The go-to tool is Doench 2016 Rule Set 2 / Azimuth, which scores a guide straight from its sequence [1], built on the earlier position-and-PAM rules of Rule Set 1 [2].

THE SEED IS KING

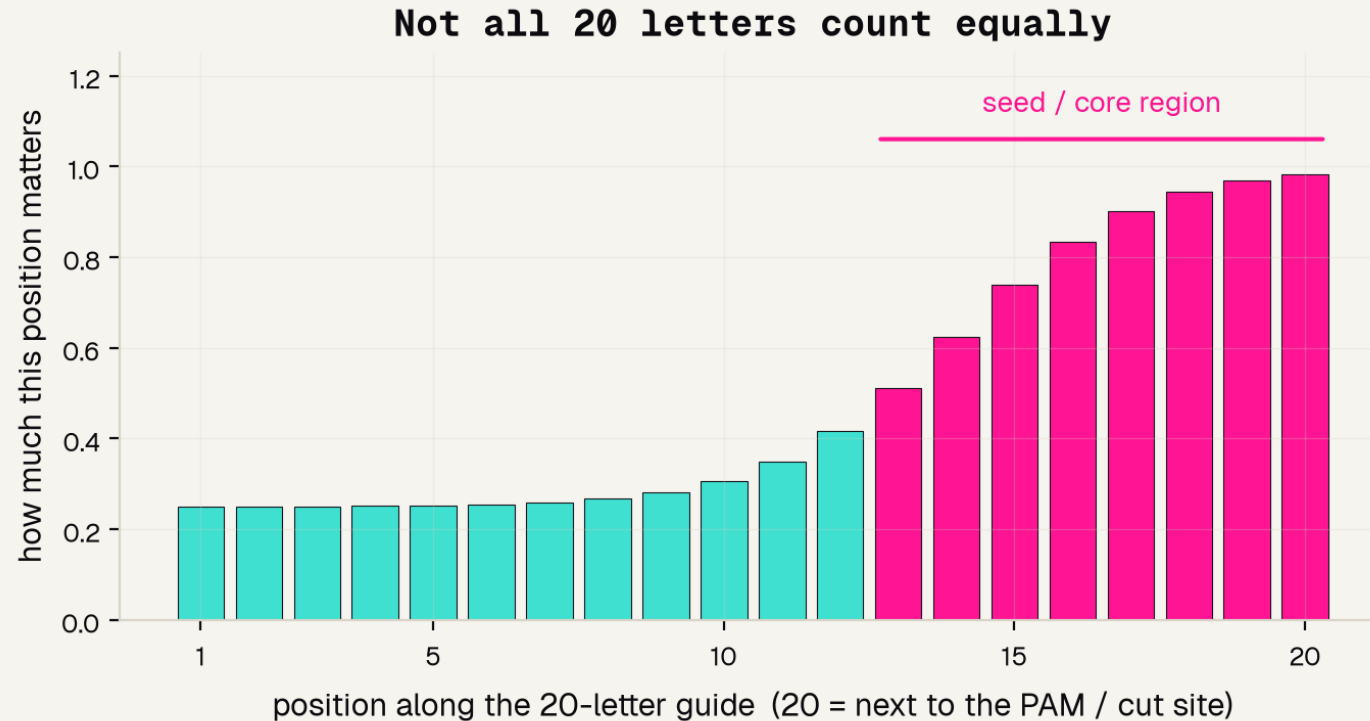
Cas9 grabs a landmark called the PAM, then checks the handful of letters right next to it -- the seed. Guides that mismatch there usually fail to cut [3][4].

NEURAL NETS PUSH FURTHER

Newer tools like DeepCRISPR learn the same job with a neural network and far more data, beating hand-built rules [5][6].

The seed region matters most

Not all 20 letters count the same. The letters nearest the PAM -- the seed, roughly positions 13 to 20 -- carry far more of the signal about whether a guide will cut.



READ THE SHAPE

The bars stay low across the far end of the guide, then rise sharply toward the PAM. That rising block is the seed, or core region.

Keep this picture in mind. With no biology lessons at all, our own model is about to rediscover this exact shape.

adapted from Zheng et al. 2017 / Doench et al. 2016

METHODS

The question we are asking

We use the exact dataset behind the famous Rule Set 2 model [1]: real guides that scientists actually measured in the lab. Our job is to predict, from the 20 letters alone, whether a guide is a strong cutter.

5,310

real 20-letter guides, each with a measured cutting score from 0 to 1

17

different human genes, so the model must learn sequence rules, not memorize one gene

~20%

of guides cut well, so a lazy "always low" guesser looks 80 percent accurate and is useless

Turning letters into numbers

A computer cannot do math on the letter A, so we turn each guide into numbers -- and how we do that is the single most important choice in the whole project. We tried two ways and let the results decide.

ONE-HOT (keeps the order)

Four on/off switches per position, so the model always knows which letter sits where. That is 20 positions times 4 = 80 numbers per guide.

K-MER (ignores the order)

Just counts how many of each letter and letter-pair appear, throwing the position away. Only 21 numbers per guide.

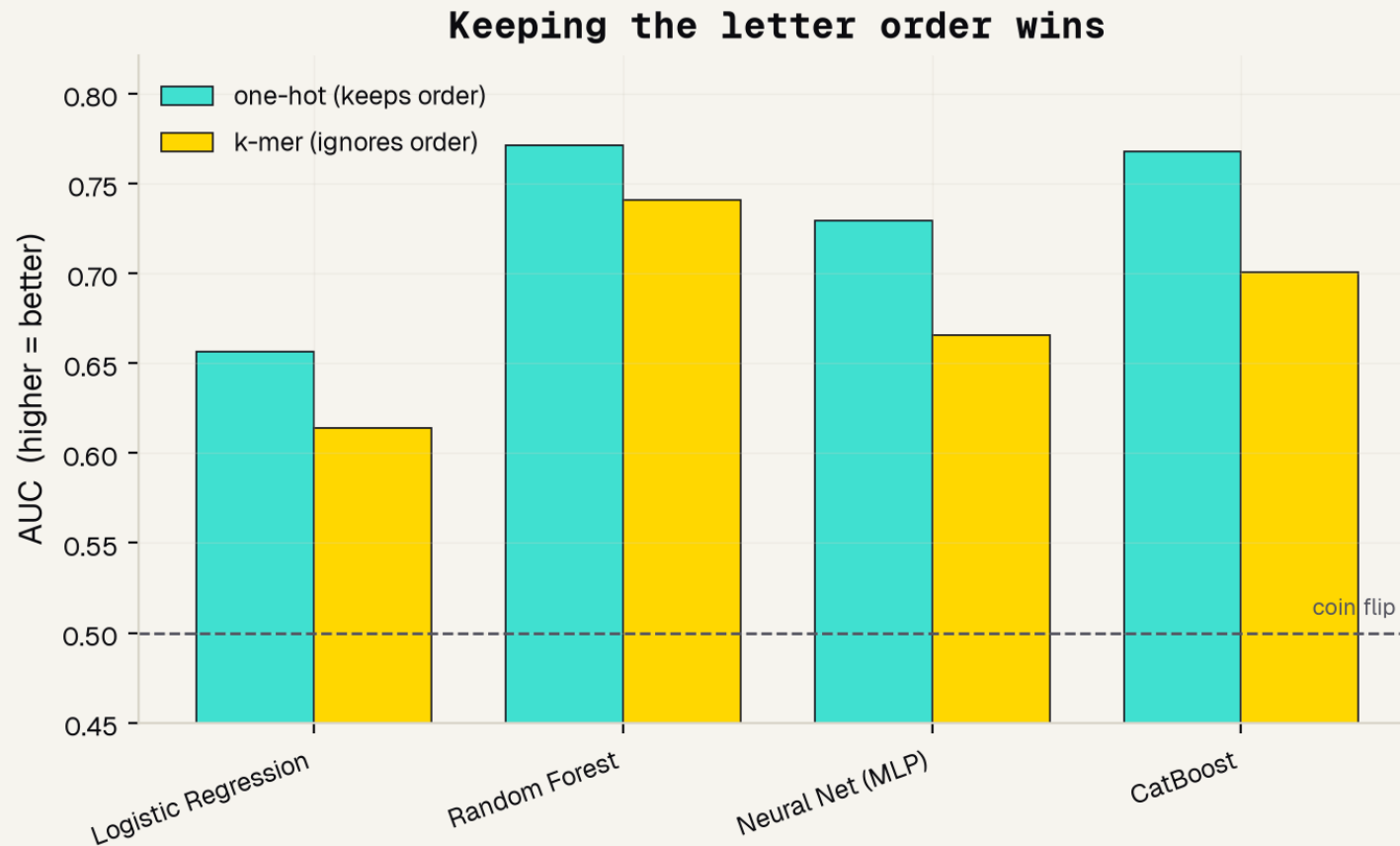
LISTEN vs SILENT

Same letters, different words. k-mer cannot tell them apart, but one-hot can -- and since biology cares exactly WHERE a letter sits (the seed!), one-hot should win.

RESULTS

Keeping the letter order wins

We trained the same four models on each way of making numbers and compared them. Every single model did better with one-hot than with k-mer.



THE REAL LESSON

The one-hot bars (turquoise) top the k-mer bars (gold) for every one of the four models. How you prepared the data mattered more than which fancy model you picked.

0.77

best AUC -- Random Forest on one-hot

Why we grade with AUC, not accuracy

Because only about 20 percent of guides cut well, plain accuracy is easy to fake. AUC asks a fairer question.

80%

accuracy of a lazy model that ALWAYS guesses "low" -- and never finds a single good guide. Accuracy is fooled by the imbalance.

0.77

AUC of our best model. AUC asks a fairer question: does it rank GOOD guides above bad ones? 0.5 is a coin flip, 1.0 is perfect.

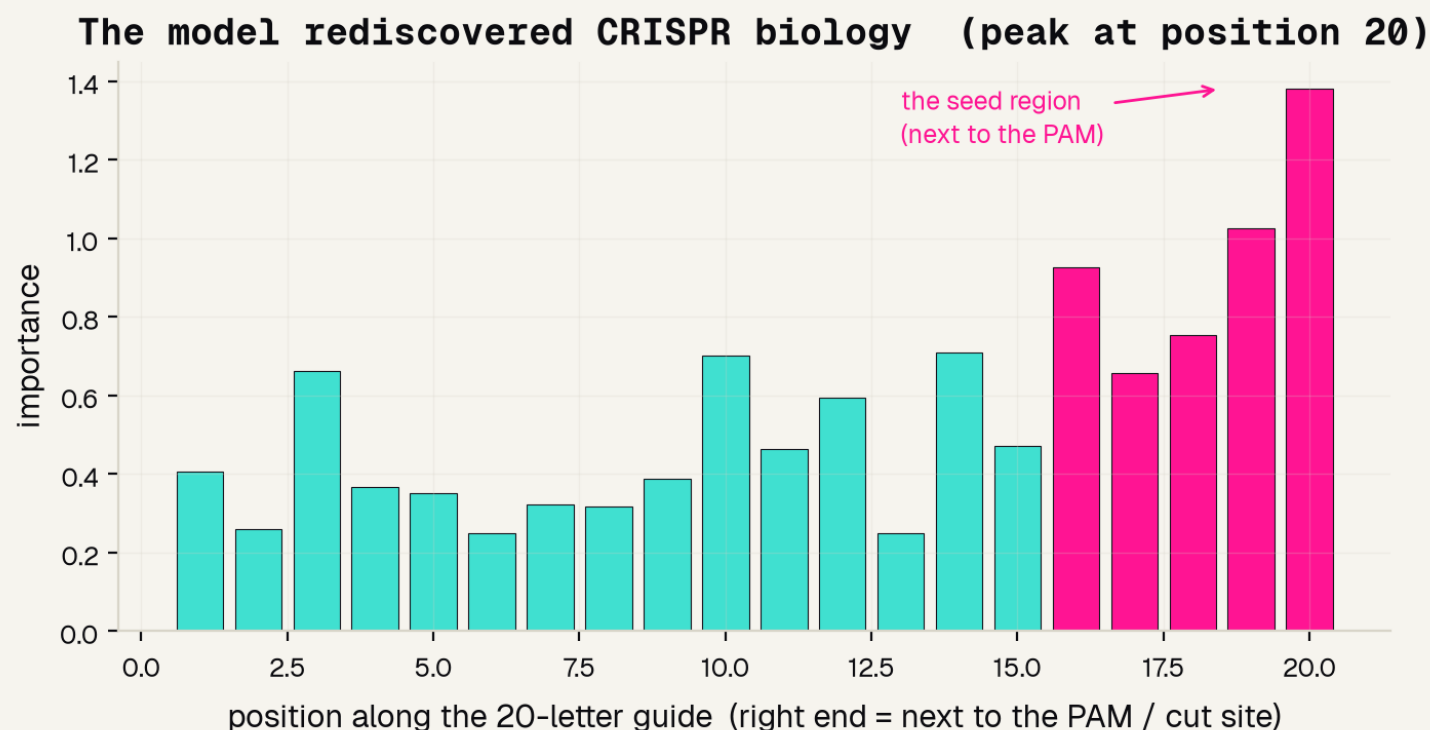
The model rediscovered the seed

Now the payoff. We asked a simple, explainable model which positions along the guide it leaned on. Nobody ever told it about seeds.

WHAT IT FOUND

Its importance climbs toward the right end of the guide and peaks at position 20, right next to the PAM. That is the seed region -- exactly where the biology said the action is.

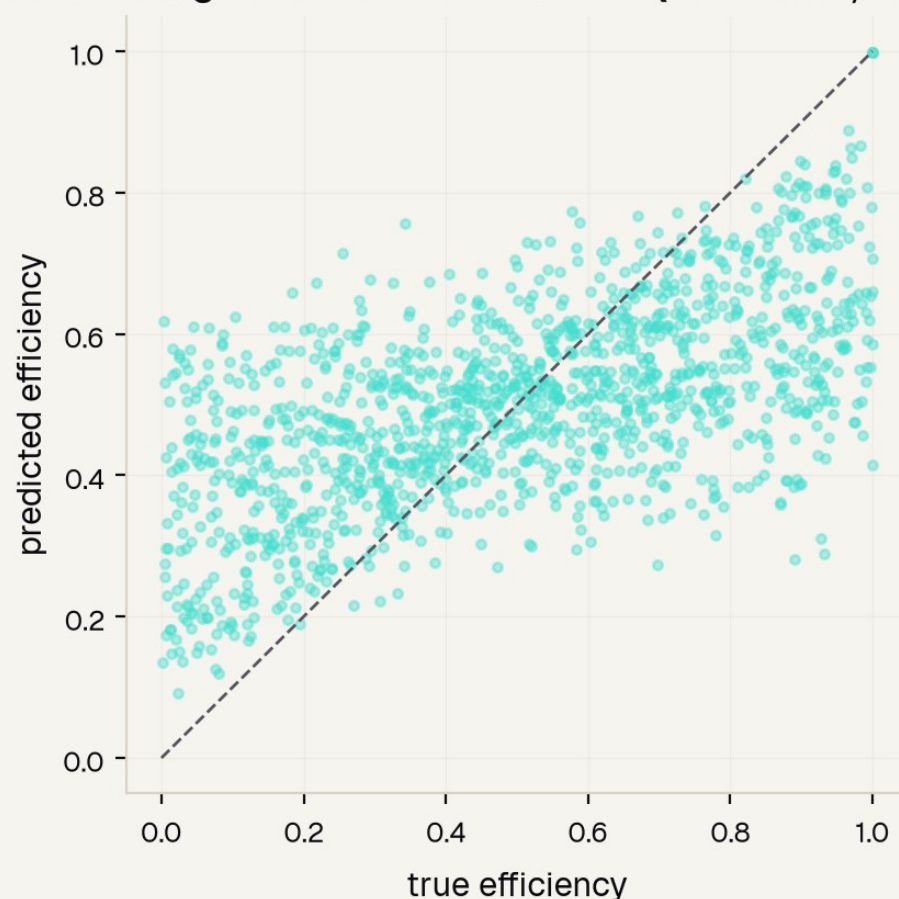
The model learned real CRISPR biology from nothing but sequences and cut-scores. A model that agrees with known science is one people will trust.



Predicting the exact score

High versus low throws away detail, so we also tried to predict the exact 0-to-1 cutting score. R-squared says how much of the real variation we explained.

Predicting the exact score (R^2 0.36, r 0.61)



0.36

R-squared: the share of the ups-and-downs the model explained (0 = guessing the average, 1 = perfect).

0.61

correlation between predicted and true efficiency.

Real signal, but far from perfect -- and that is honest. The pro tools scatter the same way [1][5][6].

BEING HONEST

What this toy model can and cannot do

A good project names its own limits out loud. Ours is a demonstration of an idea, not a design tool -- and here is exactly where the line sits.

IT CAN

Show that a guide's sequence really carries signal, and that the seed region matters -- on the very same data the professionals use [1].

IT CANNOT MATCH THE PROS

It will not beat a production tool, and it ignores the cell-type and chromatin effects that change cutting inside a living cell [7].

IT CANNOT JUDGE SAFETY

It says nothing about off-target risk -- whether the guide also cuts the wrong places elsewhere in the genome [8].

References

The eight papers behind this project -- from the plain-English CRISPR reviews to the models we measured ourselves against.

FOUNDATIONS

- [1] Doench et al. 2016, Nat Biotechnol -- Rule Set 2 / Azimuth; source of our dataset.
- [2] Doench et al. 2014, Nat Biotechnol -- Rule Set 1; position + PAM rules.
- [3] Hsu, Lander & Zhang 2014, Cell -- landmark plain-English CRISPR review.
- [4] Zheng et al. 2017, Sci Rep -- direct evidence the seed / core matters most.

MODERN MODELS & REVIEWS

- [5] Chuai et al. 2018, Genome Biol -- DeepCRISPR, a CNN guide-efficiency model.
- [6] Kim et al. 2019, Sci Adv -- DeepSpCas9, trained on 12,832 sites.
- [7] Konstantakos et al. 2022, NAR -- survey comparing guide predictors.
- [8] Abbaszadeh & Shahlai 2025, arXiv -- explainable AI + off-target safety.

The honest bottom line

A model you can explain is a model people trust.

Use this project to build intuition -- then use a validated tool and the lab to actually choose a guide.

Thank You

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